

Scalability & Future Plan

Date	
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only a limited number of users at a time.	Performance may decrease with many simultaneous users.	High
2	Uses a static dataset for prediction.	Prediction accuracy may reduce as new data becomes available.	High
3	Limited security and authentication features.	Unauthorized users may access the application.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for small-scale usage.	Deploy on cloud platforms (AWS/Azure/GCP) with load balancing to support more users.
2	Data Storage	Uses a local dataset.	Store data in a cloud database such as MySQL, PostgreSQL, or MongoDB.
3	Performance	Single-machine processing.	Optimize the ML model, implement caching, and improve API response time.
4	Security	Basic input validation.	Add user authentication, HTTPS, role-based access, and secure API endpoints.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate real-time HDI data from trusted public data sources.	3–6 Months	More accurate and up-to-date predictions.
Phase 3	Develop a mobile-friendly web application with interactive dashboards.	6–9 Months	Improved accessibility and user experience.
Phase 4	Add AI-powered recommendations to suggest ways to improve HDI indicators.	9–12 Months	Better decision support for policymakers and researchers.